

1.16 Major Risks

Risk	Impact	Solution
Duplicate bed allotment	High	Use database constraints and availability checks
Incorrect student data	Medium	Add validation and staff verification
Unauthorized access	High	Use role-based access control
Data loss	High	Maintain regular backups
Delayed development	Medium	Divide work into small increments
Poor usability	Medium	Test interfaces with users
Incomplete testing	High	Prepare test cases before deployment

2.4 Non-Functional Requirements

2.4.1 Performance

The system should load common pages quickly. Search results should be displayed within a reasonable time under normal usage.

2.4.2 Security

The system shall protect sensitive records through:

- Secure login
- Password hashing
- Role-based access
- Input validation
- Session management
- Database backup
- Audit logs

2.4.3 Usability

The system should be easy to use. Forms should contain clear labels, validation messages and simple navigation.

2.4.4 Reliability

The system should prevent duplicate allotments and maintain accurate records.

2.4.5 Scalability

The system should support additional hostels, rooms, students and modules in future versions.

2.4.6 Maintainability

The code should be modular and documented so developers can update the system easily.

2.4.7 Availability

The system should remain available during important application and renewal periods.

2.4.8 Responsiveness

The system should work on desktop computers, laptops, tablets and mobile phones.

2.6 Constraints

The system will operate under the following constraints:

1. Hostel seats depend on availability.
 2. Only authorized users can approve applications.
 3. A student cannot have more than one active allotment.
 4. A bed cannot be assigned to two students at the same time.
 5. Internet access is required for online use.
 6. Student data must remain private.
 7. Official decisions remain the responsibility of university staff.
 8. The database must preserve allotment history.
 9. Fee records must only be updated by authorized staff.
 10. The interface must support shared-room accommodation.
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